

# Helix OTA — Master Comprehensive Test-Coverage + Anti-Bluff Build-Out Plan

**Revision: 1 Last modified: 2026-06-23T13:20:00Z Goal (operator mandate, 2026-06-23):** genuine 100% real coverage by every supported test type (unit, integration, e2e, full-automation, stress, security, benchmark, chaos, ui/ux, perf/scaling) + Challenges (Challenges submodule) + HelixQA suites — all exercising **the real production-ready system running on the configured cluster**, every PASS backed by rock-solid captured physical evidence, comprehensive anti-bluff everywhere, no false positives. **Honesty (§11.4.6):** this is a multi-phase program, not a one-shot. Sources: the three audit docs in this directory (test-coverage-audit / cluster-and-system / challenges-helixqa-antibluff), each FACT-cited.

## Current state (FACT, from the three audits)

- **Go layer is strong + genuinely anti-bluff** — measured `go test -cover` (health 100% ... `ota-validator/rollout` 100%), `-race/vet` clean, and **mutation-proven**: flipping the `OTA ed25519` signature check to `always-accept KILLED` 11 tests. No bluff tests in the sampled set.
- **The pgx/Postgres Repository is fully implemented** (real-DB is one env var away); `store/rollout` default coverage is understated only because the `-tags` integration suite wasn't run.
- **"Configured cluster" is aspirational for OTA** — one live podman host (thinker) runs the HelixTrack stack, NOT `ota-server`; no multi-node/k8s. Primary blocker: no combined `ota-server+postgres` compose stack for one-command real-system boot.
- **Challenges + HelixQA are powerful but not wired against the OTA server** (shell-dispatch bank only; the `Go cmd/helixqa` orchestrator + vision toolchain are uninstalled against OTA).
- **Anti-bluff pattern is correct but partial** — applied to ~2–3 of ~17 pre-build gates; `ab_pass_with_evidence` per-assertion helper unimplemented in `helix_ota`; 14 pre-build gates are bare anchor-presence greps.

## Phase 0 — foundational enablers (highest leverage; unblock everything)

| ID        | Deliverable                                                                                                                          | Unblocks                                       | Evidence on done               |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|
| F-CLUSTER | <code>server/deploy/system.compose.yml</code> ( <code>ota-server</code> + <code>postgres</code> , real envs + <code>/readyz</code> ) | <code>integration/e2e/full-auto/stress/</code> | the harness boots the stack, / |

| ID                  | Deliverable                                                                                                                                                     | Unblocks                                                               | Evidence on done                                          |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------|
| F-<br>ANTIBLUFF-LIB | healthcheck) + a Go boot harness (mirror postgres_integration_test.go) via the containers brick pkg/ {boot, compose, health} → returns the live base URL        | security/chaos against the REAL system (real HTTP + real DB, §11.4.27) | readyz→200, captured                                      |
|                     | tests/lib/anti_bluff.sh — ab_pass_with_evidence <desc> <path> (refuses PASS unless path exists+non-empty), ab_skip_with_reason <closed-set> (§11.4.69)          | mechanical per-assertion no-evidence-no-PASS across all tests/         | self-test: PASS-without-evidence→FAIL, with-evidence→PASS |
| F-METAGATES         | tests/meta/meta_test_<gate>.sh per functional gate (mutate→FAIL→restore→PASS) + tests/meta/run_all.sh into pre-build; implement CM-SEMGREP-WIRED (no fail-open) | every gate becomes bluff-proof (§1.1)                                  | each meta-test: mutated→gate FAILs, restored→PASSes       |
| F-CLUSTER-DEPLOY    | a deploy/ota-system/ compose + distribute_stack.sh path that deploys ota-server+postgres to thinker = the real “cluster” target                                 | “real system on the configured cluster” testing                        | podman ps (healthy) + / readyz→200 on thinker, captured   |

## Phase 1 — per-test-type real coverage to 100% (risk-descending, §11.4.132)

Each closes with: real run against F-CLUSTER’s live system + captured evidence (§11.4.5/§11.4.69) + a paired §1.1 mutation routed through F-ANTIBLUFF-LIB.

1. **Integration (pgx/Postgres)** — run `-tags integration`, capture store/rollout real coverage → from UNCONFIRMED to measured. (*in progress*)
2. **e2e / full-automation** — tests/e2e/\* against the live system stack (real HTTP+DB), aggregated into one mechanical runner; cover every flow (auth, device-register, artifact upload+signature-verify, rollout staged+halt, recall, telemetry).
3. **Security / DDoS** — extend `security_probes.sh` (authz/injection/JWT-tamper/the request-supplied-key-ignored trust boundary) + rate-limit saturation + go test -fuzz on upload/telemetry parsers.
4. **Stress / chaos (§11.4.85)** — HTTP load with p50/p95/p99 (wire tools/ loadtest); chaos: malformed/oversized artifact, replay, concurrent rollout, mid-write SIGKILL, postgres-down/network-fault recovery.

5. **Benchmark** — benchstat baseline registry for the 7 existing benchmarks + regression guard.
6. **Android (instrumentation/ui)** — Gradle/JaCoCo coverage for the two JVM bricks + on-device A/B (Cuttlefish cvd — already proven Tier-2, or RK3588) with captured evidence.
7. **cmd/\* smoke** — boot each main.

## Phase 2 — Challenges + HelixQA against the real system

- **AB-G4** new OTA challenges (each real dispatch + evidence artifact + paired mutation): bad-signature-rejected, request-key-ignored, telemetry-schema, rollout-staged+halt, chaos.
- **AB-G5** wire the Go cmd/helixqa orchestrator (native crash/ANR/step-validation/evidence) against the live OTA system; bridge UI recordings through HelixQA vision (helixqa-text/recvalidate/recording-analyzer) for read-the-screen PASS (§11.4.27/§11.4.160).
- Run a full HelixQA autonomous session over the helix\_ota bank against F-CLUSTER's live system; capture the evidence ledger.

## Phase 3 — anti-bluff everywhere (AB-G1..G3, woven through every phase)

Every test type + Challenge + HelixQA result MUST: cite a captured-evidence path (F-ANTIBLUFF-LIB refuses PASS otherwise); carry a paired §1.1 mutation (F-METAGATES); the §11.4.165 independent-verifier on the batch; the §11.4.163 media-validation pipeline for any recorded artifact. No gate without its mutation; no PASS without its evidence path.

## Coverage ledger (live — updated as phases close)

| Test type         | Current real coverage                               | Target                                               | Status                                               |
|-------------------|-----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| unit (Go)         | server 47.9–100%<br>per pkg; ota-* 98–100%          | 100% real                                            | strong; store/<br>rollout pending<br>integration run |
| integration (pgx) | UNCONFIRMED<br>(suite unrun)                        | measured 100%<br>of pgx paths                        | <b>in progress</b>                                   |
| e2e / full-auto   | real shell dispatch<br>vs in-memory<br>server       | real system +<br>aggregated runner                   | F-CLUSTER<br>pending                                 |
| security/ddos     | 59 hard asserts +<br>2 rate-limit                   | + fuzz +<br>saturation + trust-<br>boundary negative | partial                                              |
| stress/chaos      | ota-* bricks have<br>it; server HTTP-<br>layer none | §11.4.85 full +<br>latency histogram                 | gap                                                  |

| Test type                  | Current real coverage             | Target                                   | Status                               |
|----------------------------|-----------------------------------|------------------------------------------|--------------------------------------|
| benchmark                  | 7 exist, no baseline              | benchstat registry                       | gap                                  |
| Android instrumentation/ui | 74 @Test, JaCoCo %<br>UNCONFIRMED | measured + on-device A/B                 | gap                                  |
| Challenges                 | 15 HOTA banks (shell-dispatch)    | + OTA holes + paired mutations           | partial                              |
| HelixQA                    | bank-runner gates only            | Go orchestrator + vision against live    | gap                                  |
| anti-bluff gates           | 2–3 of ~17 bluff-proof            | all bluff-proof + per-assertion evidence | <b>AB-G1/G2<br/>highest-leverage</b> |

## Execution discipline

Subagent-driven (§11.4.70), parallel background streams (§11.4.103, paced for the API throttle), each deliverable producing rock-solid captured evidence under docs/qa/<run-id>/ (§11.4.83), honest §11.4.6 (measured numbers as FACT, unrun as UNCONFIRMED, blocked-with-reason never faked). Phase 0 first (it unblocks everything); then Phase 1 risk-descending; Phases 2–3 woven in.